Quiroga-style QPT campaign summary

Campaign: 2–3 October 2026

Code revision: `d62fcf192f27d8fb207ce61a1c466fc5059a7e1f` (`stochastic-frames`)

> **Headline result.** On the four-qubit full design, the printed Quiroga adaptive update and FRAMES do not pass the declared recovery gate at the common 2,000-effective-row-epoch budget. An exploratory Armijo run does recover all four seeds at 20,000 steps, showing that the inverse problem itself is recoverable.

Validation and provenance

- New campaign runner and hardening tests: **88 targeted tests passed**.
- Full repository suite: **517 passed, 55 skipped**.
- Ruche smoke job `1992000`: completed, exit 0, empty stderr.
- Ruche arrays `1992001`, `1992028`, `1992340`, `1992357`, `1992362`, `1992370`, and `1992462`: every task completed with exit 0.
- The 248 synchronized result, script, and log files have identical SHA-256 manifests on Ruche and Elite.
- Each JSON/NPZ result records row, observation, initialization and source hashes; software versions; schedules; optimizer domain; timing; and failure status. Failed numerical trajectories were retained.

Two-qubit calibration gate

Artifacts: `/home/tony/qpt_quiroga_campaign_n2_20261002_01`

Results at 2,000 effective row epochs on calibration seed 0:

Design	Method	Fidelity	RMSE	Normalized TP
Full 128	Armijo FGD	0.999999999951	1.94e-7	9.61e-8
Full 128	Printed adaFGD	0.329718	4.52e-2	4.10e-4
Full 128	FRAMES	0.946621	7.33e-3	1.74e-2
Subset 96	Armijo FGD	0.999999999777	4.22e-7	1.35e-7
Subset 96	Printed adaFGD	0.214868	6.15e-2	5.70e-1
Subset 96	FRAMES	0.998182	1.60e-3	7.10e-3

Full-row Armijo also recovered on held-out seeds 1–3. The 96-row problem was seed-sensitive: Armijo recovered

on seeds 0, 2, and 3 but not seed 1. Printed adaFGD was poor on all held-out seeds and overflowed on full-row seed 3.

Four-qubit full-design results

The design has **8,192 rows**. The shared initialization fidelity on seed 0 was **0.0158792**.

Predeclared coarse grid

- **Printed adaFGD:** 9 of 18 settings failed numerically. None of the finite settings improved fidelity. Roughly 94% of runtime was spent in the dense adaptive spectral-norm numerator.
- **Armijo FGD:** the original `beta ≤ 100` range was finite but did not recover within 400 or 2,000 epochs.
- **FRAMES:** `beta=100` improved fidelity to 0.03993 at 400 epochs and 0.05549 at 2,000 epochs.

Explicit boundary refinement

Because every best point lay at the weakest TP penalty, `beta=1000` and `beta=10000` were added as exploratory refinements. Results at 2,000 effective row epochs:

Method and selected setting	Fidelity	RMSE	Normalized TP
Printed adaFGD, beta=100, eta_scale=0.003	0.009211	4.52e-3	4.03e-4
Armijo FGD, beta=10000	0.425345	1.83e-3	2.10e-3
FRAMES, beta=1000	0.164678	2.45e-3	2.44e-3

The printed update still did not improve on the common initialization.

Held-out selected comparison

The selected printed-adaFGD and FRAMES settings were run on held-out seeds 1–3 at the same 2,000-effective-row-epoch budget:

Seed	Method and setting	Fidelity	RMSE	Normalized TP
1	Printed adaFGD, beta=100, eta_scale=0.003	0.009849	4.56e-3	3.56e-4
2	Printed adaFGD, beta=100, eta_scale=0.003	0.011640	4.75e-3	3.84e-4
3	Printed adaFGD, beta=100, eta_scale=0.003	0.000789	4.77e-3	3.90e-4
1	FRAMES, beta=1000	0.077822	2.94e-3	3.49e-3
2	FRAMES, beta=1000	0.012927	3.26e-3	3.47e-3

Seed	Method and setting	Fidelity	RMSE	Normalized TP
	3 FRAMES, beta=1000	0.038065	3.09e-3	3.54e-3

All six tasks in Ruche array `1992462` completed with exit 0 and empty stderr. **None passes the**

declared recovery gate. Printed adaFGD satisfies the TP threshold but fails fidelity and RMSE;

FRAMES fails all three thresholds on these held-out seeds.

Recovery check

An explicitly exploratory 20,000-step Armijo run established that the four-qubit inverse problem itself is recoverable:

Seed	Fidelity	RMSE	Normalized TP
0	0.9999999923	1.52e-7	1.10e-7
1	0.9999999157	5.13e-7	1.96e-7
2	0.9999998359	7.17e-7	2.94e-7
3	0.9999997548	8.66e-7	3.23e-7

All four pass the declared noiseless gate: fidelity at least 0.99, RMSE at most 1e-6, and normalized TP at most 1e-3. Therefore, failure of the printed adaptive rule is not evidence that the sensing instance or ordinary factor GD is broken. This 20,000-step result is exploratory and has a ten-times-larger budget than the common comparison.

Interpretation limits

This is a **Quiroga-style benchmark, not a strict reproduction**. The higher-qubit global probes and POVM are an explicit extension. POVM and optimizer constants are project choices where the paper is underspecified. The printed adaptive rule is compared as implemented rather than silently replaced by the cited line-search rule. FRAMES is constrained to an operator-norm ball, while both FGD methods are unconstrained; the archives disclose this domain difference.

The dense realization of the adaptive numerator is feasible at four qubits but dominates runtime. It is not feasible at eight qubits; the committed backend switches to a matrix-free eigensolve there, preserving the formula but not the low cost seen in small systems.

Status: No campaign jobs remain queued or running. This PDF is the presentation copy; the Markdown file is retained as its editable source.